

Computational study of Hydrogen using variational principle

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Abstract

In this work, the variational method is applied to the hydrogen atom using a finite Gaussian basis set. The generalized eigenvalue problem is solved numerically for increasing basis sizes, allowing us to estimate both the ground-state energy and wave function. The results show a clear convergence of the energy towards the exact value and illustrate the effectiveness of the variational approach in quantum-mechanical calculations.

1 Introduction

The aim of this exercise is the computational study of the Hydrogen and Helium atom. In particular we want to calculate the energy and the wave function of the ground state for these two atoms. To do that we'll use a computational code in python based on the variational method.

Variational method and its implementation

The variational method it's an elegant way to find the energy and the wave function of the ground state of a system, and lends itself very well to numerical implementation.

We consider a system with Hamiltonian $\hat{H}$ that is in a state $|\psi\rangle$. The idea of the variational method is that the energy of the system can be expressed as a functional of the state of the system, for any chosen state. Such functional is given by

$$E[\psi] = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} \quad (1)$$

Then the energy of the ground state is the minimum of such functional. This implies that the solution of the minimizing equation

$$\frac{\delta E}{\delta|\psi\rangle}[\psi] = 0 \quad (2)$$

is the wave function of the ground state $|\psi_0\rangle$. And the energy of the ground state can be find as $E_0 = E[\psi_0]$.

This method can be implemented by expressing the state of the system as a linear combination of the elements of a suitably chosen discrete basis $\{|\chi_p\rangle\}_p$.

Since the problem is solved computationally, the basis must be truncated to its first N elements, so that the state of the system is approximated as

$$|\psi\rangle = \sum_{p=1}^N C_p |\chi_p\rangle \quad (3)$$

where C_p are the expansion coefficients associated with the basis states, and they became the unknowns of the problem.

The number N determines both the computational cost and the accuracy of the approximation. A small value of N reduces the computational effort, but also limits the accuracy of the approximation. Conversely, increasing N generally provides a more accurate approximation of the wave function, at the cost of a greater computational effort.

Now, using the expression in Eq. (3) for the state of the system in the Eq. (1), the energy functional can be rewritten as

$$E(\mathbf{C}) = \frac{\sum_{p',p} C_{p'}^* C_p \langle \chi_{p'} | \hat{H} | \chi_p \rangle}{\sum_{p',p} C_{p'}^* C_p \langle \chi_{p'} | \chi_p \rangle} \quad (4)$$

where $\mathbf{C} = (C_1, \dots, C_N)$ is the N dimension vector of the coefficients of the basis. Then, applying the variational principle

$$\frac{\partial E}{\partial C_{p'}^*} = 0 \quad (5)$$

comes out the equation

$$\sum_{p=1}^N \langle \chi_{p'} | \hat{H} | \chi_p \rangle C_p = E \sum_{p=1}^N \langle \chi_{p'} | \chi_p \rangle C_p \quad (6)$$

Now, recalling that $\langle \chi_{p'} | \hat{H} | \chi_p \rangle = H_{p'p}$ and $\langle \chi_{p'} | \chi_p \rangle = S_{p'p}$ are the matrix elements of $\mathbf{H}$ and $\mathbf{S}$, the Eq. (6) can be rewritten as the matrix equation

$$\mathbf{H}\mathbf{C} = E\mathbf{S}\mathbf{C} \quad (7)$$

where the unknowns are the energy (E) and the coefficient vector $\mathbf{C}$. For a non-trivial solution to exist, we require

$$\det(\mathbf{H} - E\mathbf{S}) = 0 \quad (8)$$

from which we find the values ($E_1, \dots, E_N$) and the vectors ($\mathbf{C}_1, \dots, \mathbf{C}_N$) that correspond to the energy and the coefficients for the first N possible states of the system.

We can now summarize the computational procedure described above in a sequence of operative steps. The aim is to obtain an approximate value for the ground-state energy and wave function of a system whose Hamiltonian $\hat{H}$ is generally too complex to be solved analytically.

1. Choose a suitable finite basis

$$\{|\chi_p\rangle\}_p$$

2. Choose the value of N reminding the previous considerations

3. Calculate the Hamiltonian and overlap matrix elements

$$H_{p'p} = \langle \chi_{p'} | \hat{H} | \chi_p \rangle \quad S_{p'p} = \langle \chi_{p'} | \chi_p \rangle$$

4. Fill the matrices **H** and **S** with the elements just calculated
5. Solve the generalized eigenvalue problem to obtain the approximate energies and wave functions of the system

$$\det(\mathbf{H} - E\mathbf{S}) = 0$$

To conclude, all the steps described above can be easily implemented in Python, which will be used throughout this work.

System

The system considered is the Hydrogen atom, composed by one fixed proton and one free electron, whose position is labelled by the vector **r**.

The energy and wave function of the system can be determined by solving the Schrödinger equation

$$\left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right) \psi(\mathbf{r}) = E\psi(\mathbf{r}) \quad (9)$$

where $\hbar$ is the Plank constant, m and e are the mass and charge of the electron and ϵ_0 is the electric constant. For this simple system this equation can be analitically resolved, and the energy and the wave function are given by

$$E_0 = -\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \approx -13.6058 \text{ eV} \quad \psi_0(r) = \frac{2}{a_0^{3/2}} e^{-r/a_0} \quad (10)$$

where $a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \approx 0.52918\text{\AA}$ is the Bhor radius.

Since an analytical solution exists for this problem, a numerical treatment is not strictly necessary. However, this simple case provides a useful test case for understanding the variational method, which can then be applied to more complex problems.

2 Implementation

For the numerical implementation, we follow the steps described in the previous section and translate them into Python.

1 – Import the required libraries

We first import the libraries needed for the numerical calculations and for plotting the results.

```
1 import numpy as np
2 from scipy.linalg import eigh
3 import matplotlib.pyplot as plt
```

2 – Define the Gaussian basis

We choose the basis functions $|\chi_p\rangle$, whose representation in real space is

$$\chi_p(\mathbf{r}) = e^{-\alpha_p r^2} \quad (11)$$

The Gaussian exponents are chosen as

$$\alpha_1 = 13.00773, \quad \alpha_2 = 1.962079, \quad \alpha_3 = 0.444529, \quad \alpha_4 = 0.1219492.$$

For calculations with $N = 1, 2, 3, 4$, we use the first N basis functions. The exponents are initialized as

```
1 # Gaussian exponents
2 alpha = np.array([
3     13.00773
4     1.962079
5     0.444529
6     0.1219492
7 ])
```

3 – Construct S and H matrices

We then calculate the matrix elements S_{pq} and

$$H_{pq} = T_{pq} + A_{pq} \quad (12)$$

whose analytical expressions are

$$S_{pq} = \int d^3r e^{-\alpha_p r^2} e^{-\alpha_q r^2} = \left(\frac{\pi}{\alpha_p + \alpha_q} \right)^{3/2}, \quad (13)$$

$$T_{pq} = -\frac{1}{2} \int d^3r e^{-\alpha_p r^2} \nabla^2 e^{-\alpha_q r^2} = \frac{3\alpha_p \alpha_q \pi^{3/2}}{(\alpha_p + \alpha_q)^{5/2}}, \quad (14)$$

$$A_{pq} = - \int d^3r e^{-\alpha_p r^2} \frac{1}{r} e^{-\alpha_q r^2} = -\frac{2\pi}{\alpha_p + \alpha_q}. \quad (15)$$

These expressions are implemented as

```

1 # Number of basis functions
2 n = len(alpha)
3
4 # Initialize matrices
5 S = np.zeros((n, n))
6 T = np.zeros((n, n))
7 A = np.zeros((n, n))
8
9 # Fill the matrices
10 for p in range(n):
11     for q in range(n):
12         S[p, q] = (np.pi / (alpha[p] + alpha[q]))**(3/2)
13
14         T[p, q] = (
15             3 * alpha[p] * alpha[q] * np.pi**(3/2)
16             / (alpha[p] + alpha[q])** (5/2)
17         )
18
19         A[p, q] = -2 * np.pi / (alpha[p] + alpha[q])
20
21 # Hamiltonian
22 H = T + A

```

Step 4 – Solve the generalized eigenvalue problem

For each basis size N , we select the corresponding $N \times N$ submatrices and solve Eq. (8).

```

1 for N in range(1, n + 1):
2     # Truncated matrices
3     H_N = H[:N, :N]
4     S_N = S[:N, :N]
5
6     # Solve generalized eigenvalue problem
7     eigenvalues, eigenvectors = eigh(H_N, S_N)

```

The solution provides the eigenvalues

$$(E_1, \dots, E_N) \quad (16)$$

and the corresponding eigenvectors

$$(\mathbf{C}_1, \dots, \mathbf{C}_N). \quad (17)$$

The lowest eigenvalue gives the variational estimate of the ground-state energy, while the corresponding eigenvector contains the coefficients used to reconstruct the ground-state wave function.

3 Results

We now analyze the results obtained with the variational calculation.

The ground-state energies obtained for different values of the basis size N are reported in Tab. 1.

N	E_0 (eV)
1	374.327019
2	19.260701
3	-11.760145
4	-13.586064
Exact	-13.6057

Table 1: Variational ground-state energy as a function of the basis size N , compared with the exact hydrogen ground-state energy.

As expected from the variational principle, increasing the number of basis functions leads to a progressive decrease of the ground-state energy towards the exact value. In particular, for $N = 4$ we obtain

$$E_0 = -13.586064 \text{ eV}, \quad (18)$$

which is already very close to the exact hydrogen ground-state energy,

$$E_0^{\text{exact}} = -13.6057 \text{ eV}. \quad (19)$$

We can also analyze the corresponding ground-state wave functions. The variational wave functions obtained for different values of N are shown in Fig. 1, together with the exact hydrogen 1s wave function.

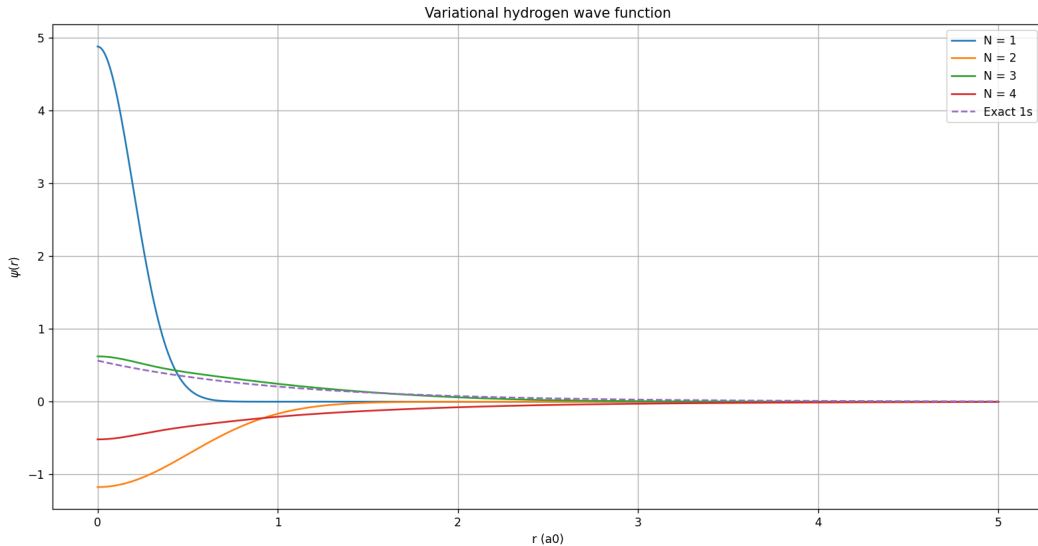


Figure 1: Variational hydrogen ground-state wave functions for different basis sizes N , compared with the exact 1s wave function.

The quality of the approximation improves significantly as the size of the basis is increased. In particular, the result obtained with $N = 3$ already reproduces the overall

shape of the exact wave function remarkably well. The $N = 4$ wave function may appear slightly less accurate in some regions of the plot; however, its corresponding energy is closer to the exact ground-state energy. This shows that a better variational estimate of the energy does not necessarily imply a uniformly better pointwise agreement of the wave function.

In general, further increasing the number of suitably chosen basis functions is expected to provide an increasingly flexible representation of the wave function and to improve the variational description of the ground state.

4 Conclusions

In this work, the variational method was applied to the hydrogen atom using a finite Gaussian basis set. By solving the generalized eigenvalue problem, approximate values of the ground-state energy and wave function were obtained for increasing basis sizes.

The results show that increasing the number of basis functions improves the variational estimate of the ground-state energy, which progressively approaches the exact value. With four Gaussian functions, an energy of -13.586064 eV was obtained, in very good agreement with the exact value of -13.6057 eV.

The calculation therefore provides a simple and effective example of how the variational principle can be implemented numerically to approximate the properties of a quantum system.

References

- [1] J. M. Thijssen, *Computational Physics*, Cambridge University Press.